DataGolf FAQ Deep Review → VenueDNA Tier Engine Improvement Roadmap

After reviewing the full DataGolf FAQ and cross-referencing it against the live 2026 Genesis Scottish Open VenueDNA board, here is a thorough breakdown of what's available, what you're already using well, and — most importantly — what could meaningfully improve the Tier Engine going forward.

✅ What You're Already Doing Well (Confirmed by FAQ)

Your four newly implemented Trait Drivers map cleanly onto DataGolf's own theoretical framework:

Your Driver	DataGolf Equivalent	FAQ Validation
TVL SG:OTT (12mo)	SG:OTT skill rating	"Long game performance is more predictive than short game performance"
HEW BallStr. (6mo)	OTT + APP composite	Matches DG's own skill decomposition methodology
BRIE SG:APP (24mo)	SG:APP skill rating	Core of DG's course-fit model adjustments
VFR SG:ARG (6mo)	SG:ARG skill rating	Incorporated in DG's category-sum normalization process

🔴 High-Priority Additions to Investigate

1. **Approach Skill Page — Distance Bin Granularity**

DataGolf's Approach Skill page breaks shots into distance bins (50–250yd from fairway, 50–225yd from rough) with four sub-metrics per bin:

- **SG per shot** (field-strength adjusted)
- **Adjusted Proximity** (closer to pin than expected)
- **Adjusted GIR %** (relative to field GIR from that bin)
- **Good Shot %** (≥ +0.5 SG)
- **Poor Shot Avoidance %** (not losing ≥ 0.5 SG)

VenueDNA Application: Renaissance's primary scoring zone is APP 150–200yd (30% weight in your Trait Profile). Your current BRIE(APP) uses the 24-month aggregate. You could add a **distance-bin-specific sub-driver** — specifically the **150–200yd fairway bin SG + Poor Shot Avoidance** — as a sharper Renaissance-specific signal, since avoiding the blow-up approach on tight fescue corridors is as important as proximity gain.

Proposed new Trait Driver:

> `BRIE-Z SG:APP 150-200 (Poor Shot Avoidance %)` — 6-month, sourced from Approach Skill page distance bins. Differentiates elite long-iron floor vs. ceiling at Renaissance.

2. **Course Fit Radar — Predictive Skill Weight at Renaissance**

The Course Fit tool runs a random effects regression indicating which specific attributes (driving distance, driving accuracy, OTT, APP, ARG, PUTT) are *statistically predictive* at each course. This is distinct from what skills players have — it tells you what Renaissance *rewards*.

VenueDNA Application: Your Venue Trait Weights (APP 150-200: 30%, OTT: 20%, etc.) appear to be your own regression. Cross-validating them against DataGolf's Course Fit output for Renaissance Club would either confirm your weights or surface divergence (e.g., maybe driving accuracy is under-weighted at 12%). This is a **model calibration check** you could run at season start or after each Scottish Open edition.

3. **Course Table — SG Difficulty by Shot Type**

DataGolf's Course Table reports how many strokes easier or harder each *shot type* plays at a course vs. average PGA Tour. For Renaissance specifically this would quantify:

- How much harder approach shots from rough play vs. fairway (critical for links ARG weighting)
- Par-5 OTT difficulty adjustment
- Putting difficulty on fescue vs. average

VenueDNA Application: Feed the **course-level SG Difficulty adjustment** into your VFS calculation as a *static venue modifier*, so that player SG ratings are re-anchored against Renaissance's actual shot-type difficulty rather than an average PGA Tour baseline. Your current BRIE(APP) baseline is implicitly "average PGA Tour" — Renaissance likely plays 0.1–0.2 SG harder on approaches from rough.

4. **Wave Split / Tee Time Adjustment**

DataGolf models the morning vs. afternoon wave scoring advantage explicitly — historically the morning wave plays **0.15–0.20 strokes easier Thursday** and **0.25–0.30 strokes easier Friday**. The Late-Early draw has a **0.10 stroke weeklong advantage** on average.

VenueDNA Application: At a coastal links like Renaissance, wind shifts can dramatically amplify wave splits beyond historical averages. You could add a **Tee Time / Wave Allocation** module to the Tier Engine that:

Flags players in the favorable wave (morning R1, afternoon R2 = Late-Early) with a small VFS bonus

Surfaces as a "Wave Risk" annotation on the leaderboard card

Sources from DG's pre-tournament wave split prediction (updated Tuesday when tee times drop)

5. **Skill Decomposition Page — Timing Adjustment (2024+ Model)**

Since 2024, DataGolf added a **TIMING column** to the skill decomposition page that adjusts for *when a player last played*. This is separate from the standard recency weighting — it specifically accounts for rust/sharpness based on recent competitive activity.

VenueDNA Application: Your FORM module currently uses True SG L5 vs. 12-month baseline. The DataGolf TIMING adjustment is a complementary signal — players who have played minimal competitive rounds in the 3 weeks before the Scottish Open may be structurally undervalued/overvalued by raw SG rolling windows. This could improve your Form module's signal clarity at a venue where course conditions (links bounce, links rough length) reward in-form sharpness.

6. **Pressure Performance in Live Model**

The DataGolf live model explicitly models **pressure effects** on projected 3rd/4th round skill based on leaderboard position. Players who historically underperform their skill under pressure (or who *over*-perform) receive adjusted finish probabilities.

VenueDNA Application: This is a gap in the current Tier Engine. There's no "Pressure Profile" driver currently visible. Consider adding a **pressure coefficient** sourced from DG's live model data — particularly relevant for the cut scenario model and R3/R4 Tier projections. DataGolf quantifies this and exposes it via API.

7. **Expected Wins (xWins) — Tournament-Quality Normalizer**

DataGolf's `xWins` metric calculates the probability that a given strokes-gained performance would result in a win in a *baseline* (average strength) PGA Tour event. This normalizes performance across field strengths — a Top 5 at the Scottish Open (stronger DP World field) is worth more xWins than a Top 5 at a weaker event.

VenueDNA Application: xWins can serve as a **career performance quality normalizer** for your Course History Tool. Instead of weighting past finishes by raw position, weight them by xWins earned — a T4 in a loaded field beats a T2 in a weak field. This would sharpen the Course History adjustment at Renaissance by only counting quality appearances.

8. **DG Points & DG Points Rankings — Peak-Performance Signal**

DG Points grade single performances using the formula `(1/probability) - 1`, where probability = likelihood of a "Top 5 Player" achieving that result. DG Points Rankings decay at 1.1%/week after 13 weeks of full value, zeroing at 104 weeks.

VenueDNA Application: DG Points could anchor a **Peak Performance at Venue** metric — specifically flagging players who have logged high DG Points performances at links/coastal courses within the 2-year rolling window. This adds a "ceiling event" signal that raw SG averages can obscure (a player who has one elite links performance is different from one who consistently posts mid-range SG).

🟠 Medium-Priority / Architecture Notes

9. **API Access — Automate All of This**

DataGolf has had a full API since December 2020. As a Scratch Pro subscriber you have access to model predictions endpoints. Everything above — skill ratings, approach bin data, course fit weights, wave splits, live model outputs — can be pulled programmatically rather than manually. This means the Tier Engine could auto-refresh Tuesday (when tee times drop + field is finalized) and again Thursday morning (post-R1 live model).

Key endpoints to target:

- `model-predictions` — skill estimates with course adjustments
- `skill-ratings` — category SG (OTT, APP, ARG, PUTT)
- `live-model` — in-tournament pressure-adjusted probabilities
- Course fit + course table data

10. **Dead-Heat Probability in Tier Scoring**

DataGolf's dead-heat adjusted probability (`p_dh`) accounts for the high frequency of ties in golf finish positions. For your **Cut % column** and tier probability outputs, if you're using raw finish probabilities without dead-heat adjustment, the displayed cut/make probabilities may be slightly miscalibrated. DG's finish tool already applies dead-heat rules — worth confirming your model mirrors this.

📊 Current Tier Engine — What's Firing Well (Live Board Observation)

From the live board, Scheffler's card shows:

TVL(OTT): +0.89 | HEW(BallStr): +1.76 | BRIE(APP): +1.29 | VFR(ARG): +0.56

NSI: 100.0, VFS implied elite across all four trait dimensions

The **Risk Vector** correctly identifies the course history gap (score -0.311 vs. field baseline) — this is already sophisticated and well-built

The four-driver architecture is coherent and the Analyst Intelligence brief output is clean. The additions above are truly *additive* — they don't fix a broken system, they sharpen an already strong one.

Summary Priority Stack

Priority	Addition	DataGolf Source	Effort
🔴 High	Approach Skill 150-200yd bin (Poor Shot Avoidance)	Approach Skill Page	Medium
🔴 High	Course Fit Radar validation of Trait Weights	Course Fit Tool	Low
🔴 High	Wave Split / Tee Time module	Pre-tournament model	Medium
🔴 High	API automation (Tuesday + Thursday refresh)	DataGolf API	High
🟡 Medium	Course Table SG Difficulty as VFS static modifier	Course Table	Medium
🟡 Medium	TIMING adjustment in Form module	Skill Decomp Page	Low
🟡 Medium	xWins as Course History quality weight	Expected Wins	Medium
🟢 Lower	Pressure coefficient (live R3/R4 only)	Live Model	High
🟢 Lower	DG Points peak-performance signal	DG Points	Medium
🟢 Lower	Dead-heat probability audit on Cut %	Matchup/Finish Tool	Low

The single highest-leverage move right now is the **Approach Skill distance-bin sub-driver** for 150–200yd, combined with confirming your Trait Weights against DataGolf's Course Fit regression output for Renaissance — those two together would measurably tighten the BRIE(APP) signal that's already your dominant scoring driver at 30% weight.